Francisco Javier Gutiérrez-Maturana Sánchez

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Date of birth: 03/05/1991



Work Experience

Actia July 2026 – Present
Embedded Systems Engineer

- Railway systems for high-speed trains, using the TRDP stack from TCNOpen.
- Embedded SECO HMI with an Intel GPU, and web technologies for real-time video decoding in the browser.
- Intensive use of large language models and artificial intelligence for development, and an optimized video-flow architecture for embedded equipment.
- Configuration of AXIS cameras in Python through SOAP and REST.
- Software testing with SonarQube and GTest, including coverage measurement and code-quality analysis.
- End-to-end integration tests with Robot Framework, and virtualized environments for a GitLab CI pipeline.
- Python bindings for Robot Framework keywords, and payload serialization for the virtual TRDP test environment.

Indra March 2026 – April 2026
Cryptographic Product Definition Engineer

- Analyzed fully homomorphic encryption (FHE), quantum random number generators (QRNG) and physical unclonable functions (PUF).
- Analyzed quantum random number generator (QRNG) use and entropy measurement for hardware quality, applying NIST methodologies for product quality.
- Developed a proof of concept of physical unclonable functions (PUF) for master-key generation, key derivation with a KDF, and private-key generation with unique identification.
- Developed a proof of concept of fully homomorphic encryption (FHE) for data privacy, with homomorphic processing that keeps client data private.
- Defined product specifications for advanced cryptographic systems within international defense and aerospace consortia (FCAS project).
- Studied and developed with Kubernetes as part of the company training.
- Developed a proof of concept of mTLS with the Python C-API and OpenSSL for cryptographic test environments.

Epicom S.A. June 2022 – February 2026
Systems, Networks and Cryptography Engineer

- Designed and implemented internal IPC protocols for confidential systems.
  - Developed a communication protocol using Unix Domain Sockets, implementing internal parallelization for inter-process communication (IPC).
  - Developed a signaling system for reliable communications, separating serial and parallel message processing functionalities.
- Designed and implemented HSM middleware based on the PKCS#11 standard, leveraging Botan for cryptographic operations.
  - Implemented thread-safe multi-threading with advanced signaling and resource contraints to prevent process contention during DoS scenarios.

  - Desgined a secure communication pipeline architecture between HSM modules, separating the internal key storage board from the external operations board at the hardware level.
  - Designed and implemented HSM core features with independent modules for cryptographic operations and system processing.
- Architected post-quantum IPsec on strongSwan, using IKEv2 and FrodoKEM.
  - Used SHAKE as the extendable-output function in the FrodoKEM key encapsulation and in the post-quantum signature schemes.
  - Optimized IKEv2 Security Association (SA) negotiation for resource-contrained devies, successfully resolving handshake collisions.
  - Resolved criticial reace conditions by optimizing timing windows to ensure reliable IKEv2 collision handling on production systems.
  - Conducted performance and stress testing on encrypted data streams using black-box FPGA implementations.
  - Developed a Secure Boot implementation in U-Boot, incorporating XMSS-based firmware verification for post-quantum resistance.
  - Resolved technical blockers during the device certification process, ensuring compliance with industry security standards.
  - Implemented advanced routing protocols using the FRR suite on embedded systems.
  - Developed Linux kernel drivers and customized Device Tree Sources (DTS) to enable SFP hot-plug support across I2C, phylink and gianfar drivers.
  - Developed and integrated SPI-NOR peripheral drivers for persistend data storage on embedded platforms.
  - Implemented custon `/proc` file system interfaces to manage and access persistent configuration data from user-space.
  - Future-proofed legacy systems by resolving Y2038 (Unix Epoch) compliance issues, spearheading the glibc migration from 2.28 to 2.40.
- Architected and designed optimized system frameworks to enhance performance, scalability, and long-term maintainability.
  - Authored comprehensive technical desing documentation for propietary hardware to meet rigorous security certification standards.
  - Architected standardized frameworks to unify the maintainability of legacy hardware platforms, reducing technical debt across multiple product lines.
- Implemented secure communication server using TLS v1.3 and Botan 3, leveraging a multi-threaded architecture with Boost.Asio for high-concurrency handling.
  - Evaluated and selected Open Source technologies based on long term maintainability and community support to ensure system longevity.
- Generated X.509 certificates and digital signatures with RSA, elliptic curves (ECDSA), CRYSTALS-Dilithium and Falcon, using the OpenSSL and Botan 3 cryptographic libraries.
  - Applied RSA, elliptic-curve algorithms (ECDH, ECDSA), AES and post-quantum schemes, including key encapsulation mechanisms (KEM) through Open Quantum Safe (OQS).
- Developed a modern web application using Next.js and Tailwind CSS, focusing on a responsive and high-performance user interface.
  - Researching and evaluating cutting-edge web technologies to architect modern, scalable and future-proof digital solutions.
  - Developed modern web interfaces using a component-based architecture with React and Next.js.
  - Implmented streamlined UI styling using Tailwind CSS, signifacntly reducing CSS bloat and accelerating the development workflow.
- Spearheaded infrastructure initiatives for the R&D department, focusing on scalability and high-performance development enviroments:
  - Implemented automated testing frameworks using Robot Framework to streamline validation for development teams.

  - Standardized Red Hat Enterprise Linux (RHEL) as the primary development platform, ensuring high-assurance and reliable environments for defense-grade projects.
  - Architected air-gapped development infrastructures, including local NPM registries to enable offline web development in high security enviroments.
  - Deployed offline large language models (Llama) in air-gapped environments for intensive coding, refactoring and faster development with artificial intelligence.
  - Enforced code quality standars by implementing GitLab commit restrictions and protected branch policies.
  - Leveraged Jira to manage complex workflows and strategic objectives, ensuring seamless coordination and tracking across the project lifecycle.

Capgenemi November 2021 – May 2022

System Software and Database Engineer

- **C++**
  - Development and design of relational databases SQL: Oracle.
  - Development and design of non-relational databases, NoSQL: MongoDB, Kibana.
  - Testing with C++ GMock and GTest frameworks and regression test environment with EDIFACT messages for airplane traveling agencies.
  - Development of Application Payment Service in Amadeus in fraud analysis and 3DS payment security authentication; VISA, Mastercard.
  - Use of Git.
- **Java**
  - Use of Java with Maven and Gradle.
  - Java 9 lambda programming and design patterns knowledge.
  - Use of Spring Boot framework.
  - Design of web services and API REST.
  - Use of microservices in Java and load balancing for scale web services in high density environments: Netflix framework.
  - Java Data Access to DB with ORM frameworks: Hibernate or JPA.
  - Use of Java monitors for concurrency applications.
  - Use of Git.

Teldat S.A. September 2018 – November 2021

System Network & Software Engineer

- Research about WiFi technologies WiFi 5 and WiFi 6.
- Development of embedded software with SoCs IPQ4019 and IPQ8072 Qualcomm chipsets.
- Testing with Robot Framework and CI with Jenkins and Git.
- Develop algorithms to use access point steering with 802.11k/v WiFi technologies (LBD).
- Maintenance of WiFi modules in the Linux kernel and monitoring and configuration subsystems in userspace C/C++, Qualcomm software.
- Experience in development using Linux kernel 4.4 with ioctl and netlink interfaces.
- Work with Debian distributions for embedded software in access points, from Buster to Jessie.
- Use of Docker and primitive containers with schroot for embedded Debian systems to use in virtual environments.

Sgenia Industrial January 2018 – August 2018

System Software Engineer

- Use of operating system Red Hat Enterprise Workstation 7.4.
- Testing of ITER CODAC 6.0 modules DAN, UDA and SDN.

- Hack of ITER CODAC 6.0 to use an own mass archiving system project based on DAN archiving and UDA Unified Data Access technologies on an IoT I+D development.
- Languages Bash and C/C++, Java and Python.
- Development with PostgreSQL.
- Collaboration with CIEMAT and Indra.

Eptisa Service of Engineering Home June 2017 – December 2017
Developer

- Develop a smart 4º Generation thermostat in an I+D project.
- Develop Google Cloud service client Datastore with App Engine on Spring Boot and with Maven, use of Java 8.
- Client database NoSQL in real time (Firebase) and an authentication system using Firebase Authentication with JWT.
- Working with Node.js and the framework Node-RED to implement the sensors of the thermostat with Zigbee and algorithms of automatic learning or Machine Learning.
- Use of the API 11 GMS of Google in Android with localization and Geofence services.
- Studying technologies and documentation in English language.

Electria Wind October 2016 – March 2017
Business Practice

- Debugging the task of an industrial computer PCL in a aerogenerator, with the RSLogix Adam Bradley tool.
- Development of an application in Android Studio to monitor the PLC data in the aerogenerator.

Consultants office's computers Soluciones SI
Business Practice

- Remote administration of computer equipment and backups with Acronis True Image for Windows systems with NTFS file system.
- Repair of computers and installation of operating systems and optimization.
- Documents and technical procedures inside the business.
- Resolution of technical problems and incidents concerning systems inside the business.

Academic Information

Technical System and Network Management
Instituto María Zayas y Soto Mayor, Majadahonda 2010 – 2013

Graduate in Telematic Engineering
Universidad Rey Juan Carlos de Madrid 2013 – 2017

Postgraduate in Computer Science
Universidad Rey Juan Carlos de Madrid 2017 – 2020

Curso de criptografía en el CSIC, microcredencial Europea
2025

Knowledge

Cryptography — Expert

- Expert in applied cryptography: RSA, elliptic-curve Diffie–Hellman (ECDH) and ECDSA, AES, key encapsulation mechanisms (KEM) and post-quantum cryptography with Open Quantum Safe (OQS).
- X.509 certificate generation and digital signatures with RSA, elliptic curves, CRYSTALS-Dilithium and Falcon, using the OpenSSL and Botan 3 cryptographic libraries.
- Fully homomorphic encryption (FHE), physical unclonable functions (PUF), quantum random number generators (QRNG) and mTLS. PKCS#11 HSMs, IPsec with strongSwan and IKEv2, FrodoKEM, hybrid ML-KEM (X25519/ML-KEM-768 and ML-KEM-1024), Argon2id and SHAKE.
- Secure boot in U-Boot with post-quantum XMSS signatures for firmware signing and verification.

Networks — Expert

- Expert in WiFi 5 and WiFi 6, 802.11k/v access-point steering, and the WiFi subsystem of access points.
- WiFi radio offload and hardware-bridge datapath acceleration, forwarding WiFi interfaces to Ethernet without interrupting the processor and at low latency.
- TCP/IP stack and the OSI model, dynamic routing (OSPF, BGP), static routes, switches, VLANs, STP, the link layer and multicast networks.
- Link technologies and modulations (AM, FM, PM): Ethernet, WiFi, WiMAX, SONET, ATM, radar and satellite systems.
- Railway TRDP with the TCNOpen stack, and the FRR routing suite.
- IPsec with IKEv2, TLS v1.3, UDP, TCP as a transport protocol, and network queue policies.
- FPGA offload of cryptographic datapath operations to accelerate cryptographic processing.

SoC and embedded — Expert

- Expert in system-on-chip platforms, including MIPS32 and PowerPC: assembler, memory hierarchy (stack, heap, data, text), microprocessor internals and DMA peripherals.
- Intel architectures with SSE, SSE2 and MMX, and embedded Qualcomm SoCs IPQ4019 and IPQ8072.
- Embedded Linux on access points, with Debian userspace in C/C++ on those SoCs.
- Boost.Asio with coroutines, userspace TLS v1.3, and the C++20 and C++23 standards.
- FastDDS and POSIX shared memory for video decoding and high-efficiency streams between processes.
- U-Boot, including secure boot, on embedded platforms.
- Toolchains and native compilation of the kernel and GCC for certification builds on ARM, ARM64, AMD64 and PowerPC.

Linux kernel — Expert

- Expert in the Linux kernel: process model, POSIX multi-threading, ioctl and netlink, and maintenance of in-kernel WiFi modules.
- Semaphores, mutexes and condition variables, and the POSIX library in general.
- Kernel drivers and Device Tree Sources (DTS) for I2C, phylink, gianfar and SPI-NOR, plus custom `/proc` interfaces.
- IPsec drivers for encrypted data flows processed by an FPGA, key management in the memory address space, and SPI for the UDP protocol in the Linux kernel.
- Kernel modifications for SFP interfaces with phylink on MIPS32 SoCs, for embedded environments and legacy equipment.
- Kernel upgrades from the 4.x series to the 5.x series on embedded systems.
- Kernel `printk` traces to debug operating-system errors, and kernel-panic analysis of the system stack after an unexpected crash and reboot.
- Kernel configuration with `menuconfig` and cross-compilation for multiple architectures.
- System hooks on socket buffers (`sk_buff`) and network-protocol manipulation at operating-system level in the Linux kernel.
- Optimization of ARP, IP, TCP and UDP through kernel `sysctl` configuration.
- Shell and boot environment around the kernel: Bash, System V services, SSH and tunneling.

Virtualization — Expert

- QEMU/KVM and VirtualBox hypervisors, Docker and Podman containers, and Kubernetes as a container orchestrator.
- Load balancers with Kubernetes.
- QEMU emulation of target architectures running cross-compiled operating systems, with fully isolated per-container network stacks over TUN/TAP interfaces.
- L2 network segmentation at the hypervisor for isolated guest operating systems and containers, using Linux bridges, network namespaces and cgroups for lightweight isolation.
- Virtualized integration environments on GitLab CI, with Robot Framework end-to-end tests and a virtual TRDP testbed.
- Docker and schroot for embedded Debian userspace on access points.
- Air-gapped build and runtime environments, including local package registries.

Databases — Expert

- Expert in relational databases, including PostgreSQL and PostGIS: SQL DDL and DML, schema design, table triggers, and roles for authorization. PostGIS keeps a WGS84 pose on a GiST index and a circular track history.
- Reactive PostgreSQL with libpqxx: event callbacks and JSON objects from the database engine.
- Oracle and PL/SQL. Non-relational stores: MongoDB, Redis, Cassandra, Firebase and Kibana.

Web technologies — Expert

- Server-side development with Spring Boot, JPA and Maven, and client-side HTML, CSS, the DOM, JavaScript, jQuery and Bootstrap.
- Interfaces with React, Next.js and Tailwind CSS.
- WebRTC and MediaMTX for video decoding, including RTSP cameras.
- Web testing and validation with Selenium and Robot Framework.
- Node.js and Node-RED. Service interfaces with REST and SOAP.
- HTTP/2 (nghttp2) and WSS over the same TLS session, with WebSockets pushing telemetry and real-time events to the browser.
- Browser flag tuning and embedded Chromium kiosk browsers for displaying web user interfaces.
- Embedded-browser tuning for GPU-accelerated video decoding and correct GPU offload in the browser.

Graphics — Advanced

- OpenGL 3.3 and GLSL, and Qt visualization with DAN and CUDA.
- Signal, image and sound processing with OpenCV and FFmpeg.
- GPU-accelerated H.264 video decoding in the browser for embedded HMI, using Intel GPUs and libva.

Systems and robotics — Medium

- Cybersecurity: iptables, honeypots, Kali Linux with the Metasploit framework, and forensic analysis.
- Parallel and GPU computing with OpenMPI, OpenMP, MapReduce and CUDA.
- Machine learning with metaheuristics, Keras and convolutional networks, and YOLO on a companion NPU. ROS and ROS 2 navigation and tracking, including a drone, with OpenCV and PCL.
- Software testing with SonarQube, GTest and code-coverage measurement. Python, including the C-API.

Cloud and data — Basic

- AWS: EC2 for clustering and high availability, load balancers ELB, DNS with Route 53, RDS and S3.
- Git, GitLab CI, Jenkins, SVN and Travis CI.

Multi-platform applications — Basic

- Android application development with Android Studio and Gradle.

IDEs

- OpenCode, Cursor and Visual Studio as development IDEs with an integrated AI agent.

Debugging analysis

- Debugging with gdb and Valgrind.
- Core-dump debugging to inspect memory corruption of userspace processes.
- Long-duration tests to check memory stability of the system.
- Browser tooling to detect memory problems in web applications.
- Selenium to debug the DOM and JavaScript objects as memory-debugging tests.
- System tools to measure performance and efficient hardware use: `intel_gpu_top` for GPU usage statistics, `htop` for Linux process analysis, and `/proc` statistics.
- Traces to debug data flows and software behaviour.
- Kernel `printk` traces to debug operating-system errors, and kernel-panic analysis of the system stack after an unexpected crash and reboot.

Languages

English — intermediate skills, TOEIC certification B1.2

Personal Projects

Rescue drone sensor platform — 2026 – Present

Personal project, management-sensors

- Search-and-rescue quadcopter: Holybro X500 V2, Cube Orange on ArduPilot, Here4 GNSS, Holybro H-Flow, STL-27L 360° lidar, ArduCam OV9782 global-shutter camera, Orange Pi 5 companion (RK3588S, 6 TOPS NPU) over Wi-Fi 6, and ExpressLRS as the stick link.
- Onboard video and AI: the camera stream is processed with YOLO on the Orange Pi NPU (RKNN). Those person detections feed the rescue tracking loop.
- Robotics on ROS 2 Lyrical: navigation and tracking algorithms that hold station on the person, using the camera bearing and the lidar range. The same node is the field-bus controller on eProsima Fast DDS (Fast-CDR, IDL, Micro XRCE-DDS and OpenSSL mTLS), with H.264 from FFmpeg into a POSIX shared-memory ring, in C++23.
- Quantum-safe management plane: Botan 3 with TLS v1.3 and hybrid X25519/ML-KEM-768 key exchange (also ML-KEM-1024, and Kyber and FrodoKEM policies), ES384 JWT, Argon2id, Boost.Asio. The web server speaks HTTP/2 through nghttp2 on that TLS session. Live channels are WSS: the WebSocket rides the same HTTP/2 connection, and each socket is authorized with the ES384 JWT.
- PostgreSQL and PostGIS, through libpqxx, as the source of truth and the event bus. LISTEN/NOTIFY decouples the TLS server from the ROS 2 controller. The drone pose is a WGS84 PointZ with a GiST index, and a circular history keeps the track. Triggers, sequences and database roles carry configuration, telemetry, video metadata and YOLO detections.
- Operator panel: Next.js 16, React 19 and TypeScript as a static export, Tailwind CSS 4, a MapLibre mission map, live telemetry over WSS and video over WebRTC. Three.js and React Three Fiber for the spatial view.
- Build and validation: CMake, colcon, Docker, AArch64 and ARM cross builds, QEMU, GoogleTest, Robot Framework with Selenium, Vitest and GitHub Actions.

Personal Achievements

- RoboCup participation in 2016–2017 with domestic robots, in collaboration with the León college and the URJC in Germany. Development in C++ and ROS of people tracker modules.
- Building a 3D printer.
- IEEE student member.
- University developments and open source contributions: https://github.com/jgmatu

Personal Interest

My main objective is to gain working experience and specialise in networking, robotics, computer design, computer vision or systems engineering.

I am a committed and responsible person with a high capacity for adaptation and integration in a working team. I would love to develop in the USA.

Boxing practitioner (1 year): consistent training at the Arganda del Rey Boxing School (2025), developing high physical endurance, discipline and focus.

Availability

Available to work full-time.